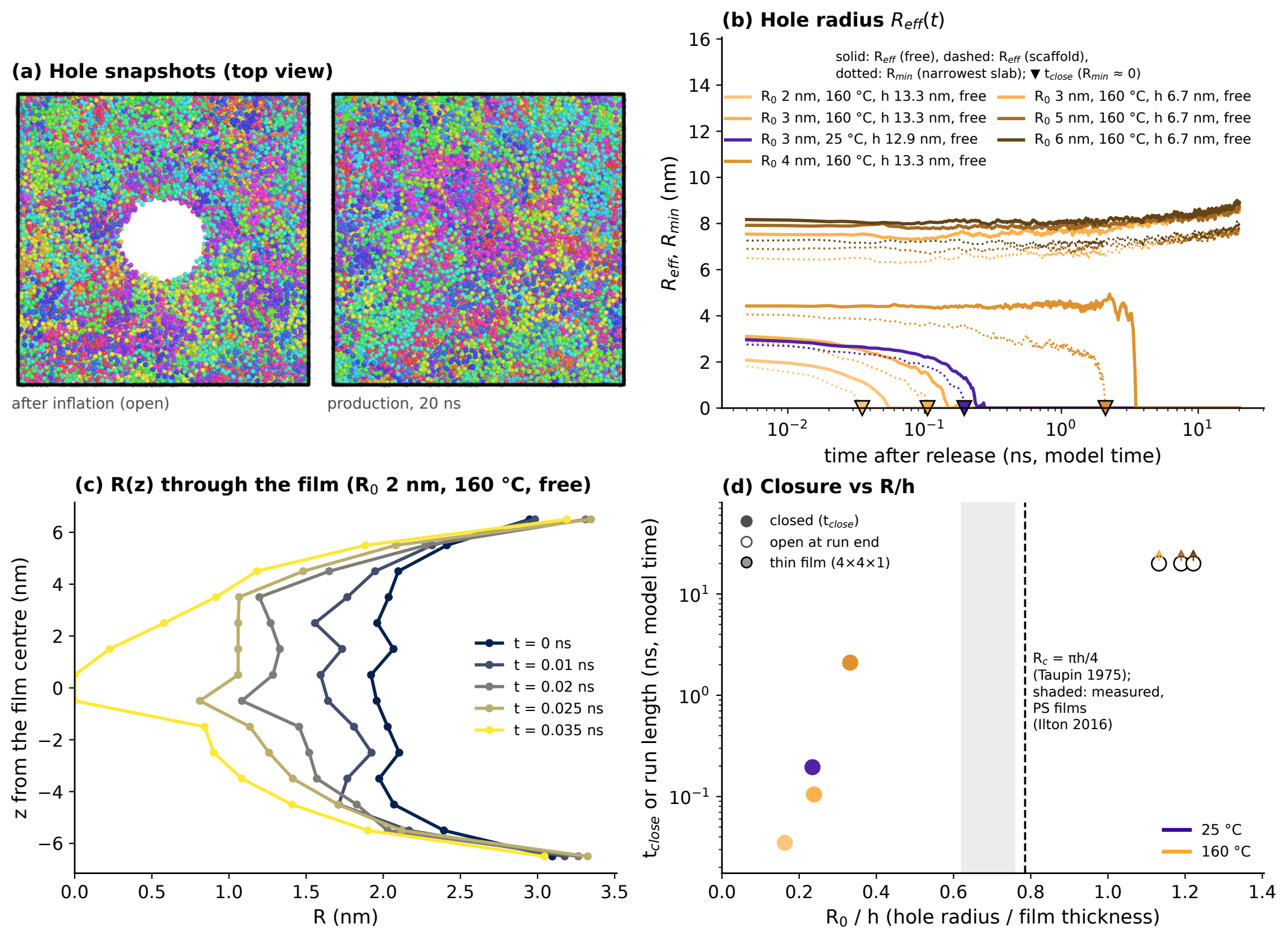

(a) Hole snapshots (top view)
after inflation (open)
production, 20 ns
(b) Hole radius $R_{eff}(t)$
solid: R_{eff} (free), dashed: R_{eff} (scaffold),
dotted: R_{min} (narrowest slab); ▼ t_{close} (R_{min} ≈ 0)
R_0 2 nm, 160 °C, h 13.3 nm, free
R_0 3 nm, 160 °C, h 13.3 nm, free
R_0 3 nm, 25 °C, h 12.9 nm, free
R_0 4 nm, 160 °C, h 13.3 nm, free
R_0 3 nm, 160 °C, h 6.7 nm, free
R_0 5 nm, 160 °C, h 6.7 nm, free
R_0 6 nm, 160 °C, h 6.7 nm, free
R_{eff}, R_{min} (nm)
time after release (ns, model time)
(c) R(z) through the film (R_0 2 nm, 160 °C, free)
z from the film centre (nm)
R (nm)
t = 0 ns
t = 0.01 ns
t = 0.02 ns
t = 0.025 ns
t = 0.035 ns
(d) Closure vs R/h
closed (t_{close})
open at run end
thin film (4×4×1)
t_{close} or run length (ns, model time)
R_0 / h (hole radius / film thickness)
R_c = πh/4
(Taupin 1975);
shaded: measured,
PS films
(Ilton 2016)
25 °C
160 °C